

Energy stores and systems

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know.

QUESTION		ANSWER	✓
1	Name the eight stores of energy.	Chemical, electrostatic, magnetic, elastic potential, nuclear, thermal, kinetic and gravitational potential	☐
2	State the law of conservation of energy.	Energy can be transferred usefully, stored or dissipated, but it cannot be created or destroyed.	☐
3	What is a closed system?	A system that no energy enters or leaves.	☐
4	Define work done.	The energy transferred when a force moves an object through a distance.	☐
5	State the standard unit of energy.	Joule (J)	☐
6	Name the store of energy associated with fuels and batteries.	The chemical energy store	☐
7	Describe the change in the chemical energy store as fuel is burned.	It decreases.	☐
8	Name the store of energy associated with changes in speed.	The kinetic energy store	☐
9	Describe the change in the kinetic energy store as an object slows down.	It decreases.	☐
10	Name the store of energy associated with changes in height.	The gravitational potential energy store	☐
11	Describe the change in the gravitational potential energy store as an object rises.	It increases.	☐
12	Name the store of energy associated with extending and compressing objects.	The elastic potential energy store	☐

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13	Describe the change in the elastic potential energy store as an object is compressed more.	It increases.	☐
14	Name the store of energy associated with changes in temperature.	The thermal energy store	☐
15	Describe the change in the thermal energy store as an object gets hotter.	It increases.	☐
16	Name the four ways energy is transferred between stores.	Mechanically, electrically, by heating and by radiation	☐
17	Name the energy transfer when a force moves an object.	Mechanical transfer	☐
18	Name the energy transfer when an object emits light or sound.	Transfer by radiation	☐
19	Name the energy transfer when a hot object warms a cooler one.	Transfer by heating	☐
20	What is a system?	An object or a group of objects.	☐
21	State what happens to energy when a system changes.	It is transferred from one store to another.	☐
22	State the equation linking kinetic energy, mass and speed.	kinetic energy = $0.5 \times \text{mass} \times \text{speed}^2$ ($E_k = \frac{1}{2} m v^2$)	☐
23	State the equation linking gravitational potential energy, mass, gravitational field strength and height.	gravitational potential energy = mass $\times$ gravitational field strength $\times$ height ($E_p = m g h$)	☐
24	State the equation linking elastic potential energy, spring constant and extension.	elastic potential energy = $0.5 \times \text{spring constant} \times \text{extension}^2$ ($E_e = \frac{1}{2} k e^2$)	☐
25	Name the two quantities that kinetic energy depends on.	Mass and speed	☐
26	Name the three quantities that gravitational potential energy depends on.	Mass, gravitational field strength and height	☐

QUESTION		ANSWER	✓
27	A ball is thrown straight up. Describe the changes in its energy stores as it rises.	Kinetic energy store decreases; gravitational potential energy store increases.	☐
28	An electric kettle is switched on. Describe the useful energy transfer.	Energy is transferred electrically to the thermal energy store of the water.	☐